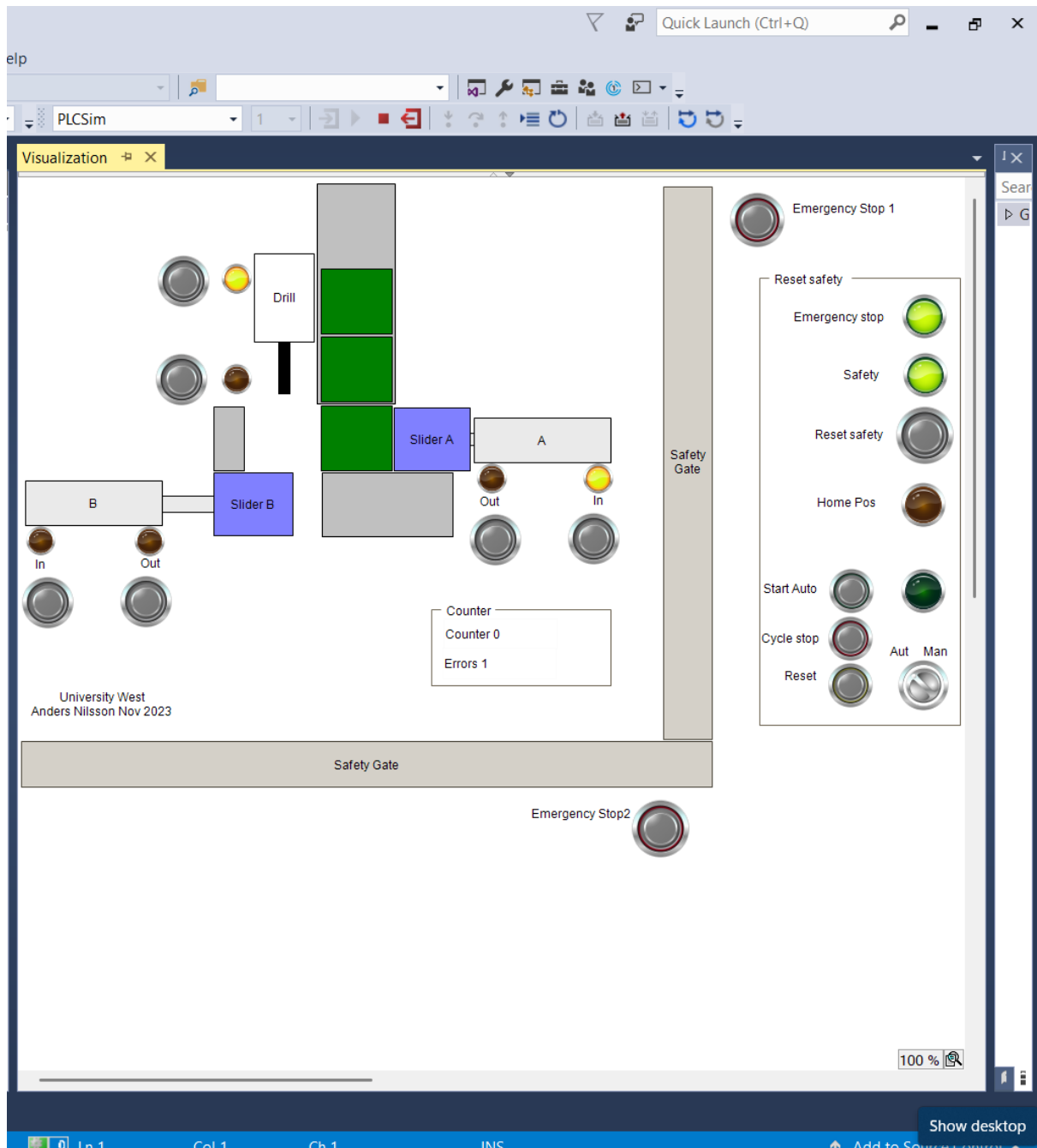




TwinCAT Project Drill - TcKaeShell

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Build 4026.20 (Loaded) Release TwinCAT OS (x64) Attach... PLCStudent

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ControlFBD [Online] ControlFBD.Sequence [Online] STATES BLINK MAIN

TwinCAT Project Drill PLCStudent.ControlFBD

Expression	Type	Value	Prepared value	Address	Comme
qxLampStarted	BOOL	FALSE		%Q*	Physical C
qxSliderAout	BOOL	FALSE		%Q*	
qxSliderAin	BOOL	FALSE		%Q*	
qxSliderBout	BOOL	TRUE		%Q*	

Sequence

1. The Emergency stop summation:
All emergency buttons must stop the machine so i add them together with AND_gate hence if even one button is pressed the summation becomes false.

ixBtn1EmStopOk TRUE & mxSumEmStopReady TRUE

ixBtn2EmStopOk TRUE

2. The reset latch:
Self_Holding_Function: The machine must not start just because someone released the emergency stop button hence they must also press the reset button.
OBS: (A + B).C

ixBtnReset FALSE OR ≥1 mxEmStopOk TRUE

mxEmStopOk TRUE AND mxSumEmStopReady TRUE mxEmStopOk TRUE

Visualization

Drill

Slider A

Slider B

Counter
Counter 0
Errors 7

Safety Gate

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ControlFBD [Online] ControlFBD.Sequence [Online] STATES BLINK MAIN

TwinCAT Project Drill PLCStudent.ControlFBD

Expression	Type	Value	Prepared value	Address	Comme
fbBlink	FB				
T#500MS OnTime	OUT	mxBlink FALSE			
T#500MS OffTime					

3. Initializing the blink:
Call the blink_function (To generate a periodic On/Off signal)

4. Indication logic:
This network will controll how the lamp behaves based on the states we just programmed.
OBS: (A + B).C

mxSumEmStopReady TRUE AND mxBlink FALSE ≥1 qxLampEmStopOk TRUE

mxEmStopOk TRUE

5. Safety stop logic (for Sensors):

ixSafety1StopOk TRUE AND ixSafety2StopOk TRUE mxSumSafetyReady TRUE

Visualization

Drill

Slider A

Slider B

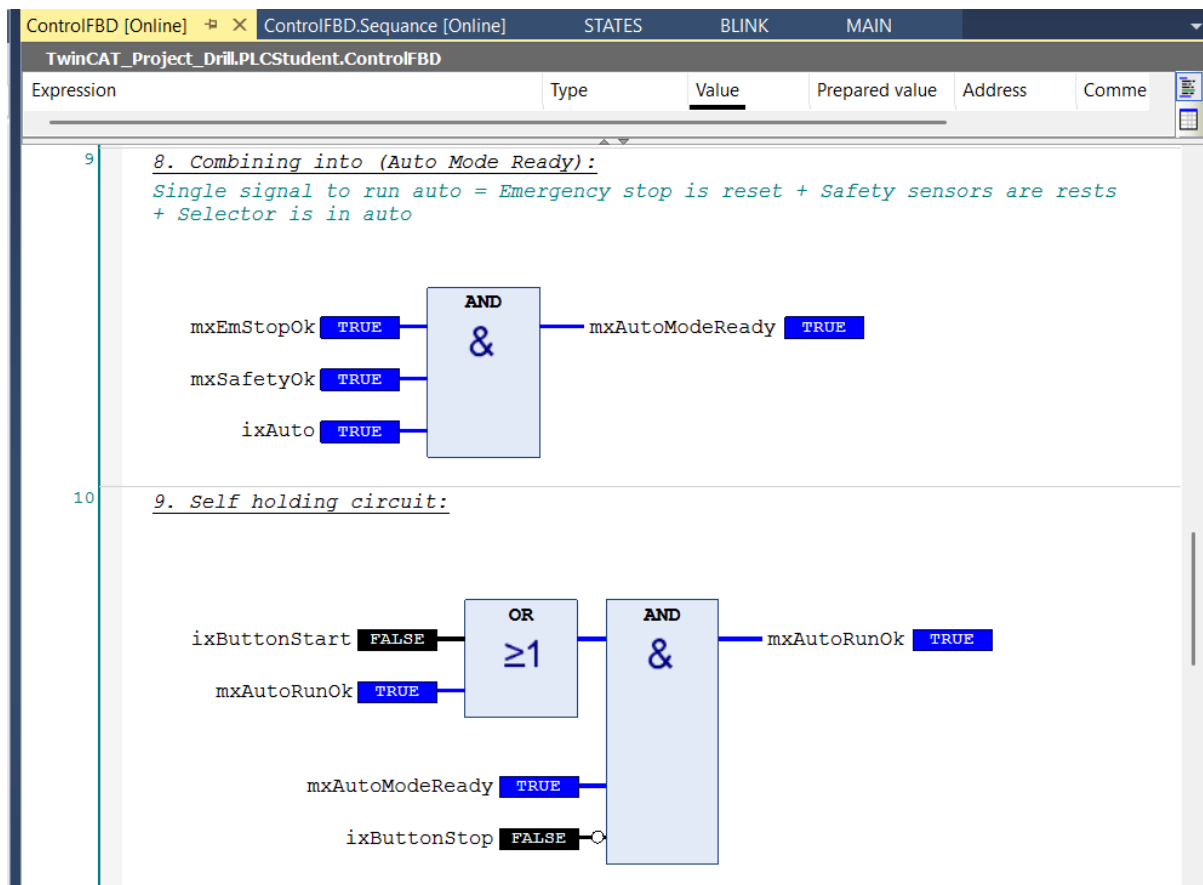
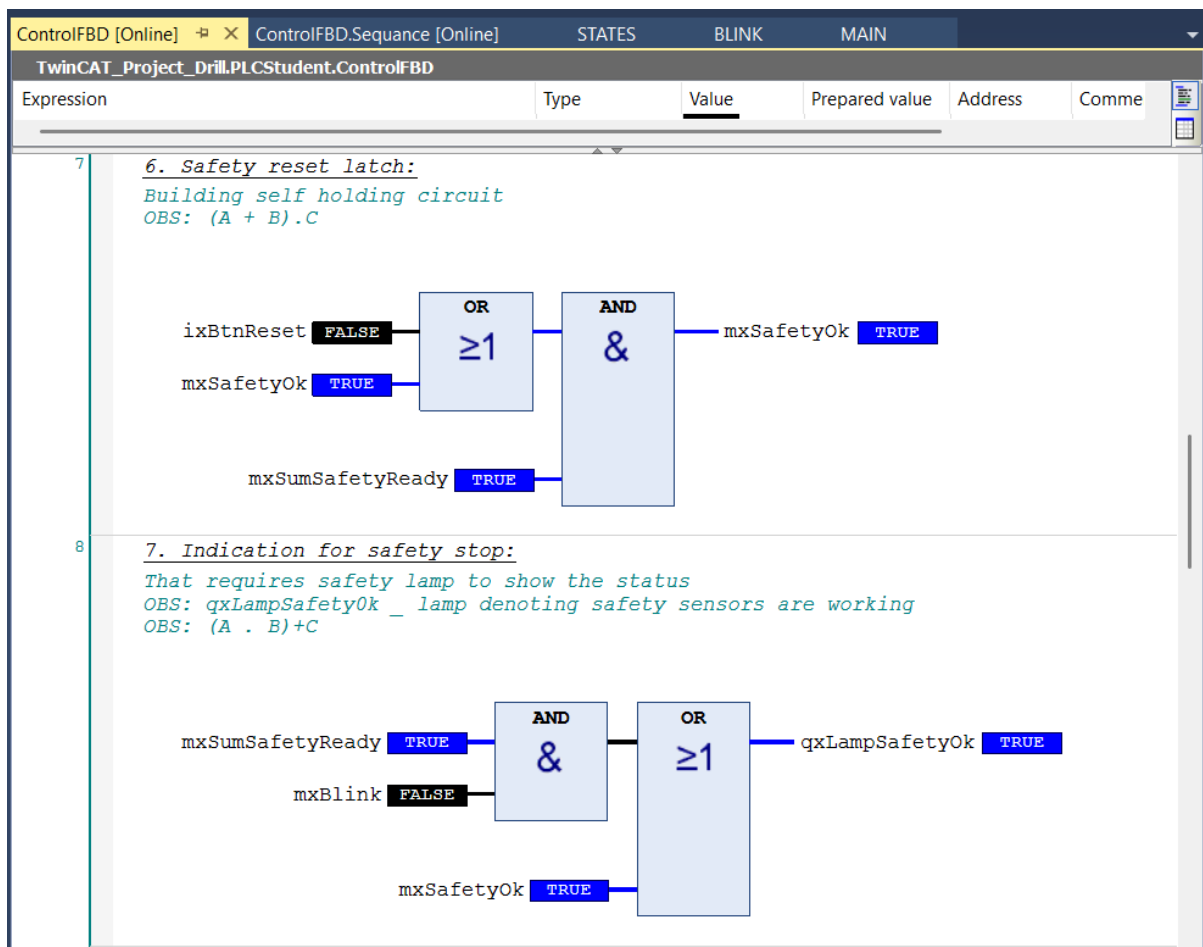
Counter
Counter 0
Errors 11

Safety Gate

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ControlFBD [Online] ControlFBD.Sequence [Online] STATES BLINK MAIN

TwinCAT_Project_Drill.PLCStudent.ControlFBD

Expression	Type	Value	Prepared value	Address	Comme
11	<u>10. Defining (Home Position):</u> The machine must be at home to start hence we must check all sensors for all moving parts.				
12	<u>11. Auto Run Latch:</u> Self holding function is the start command for the sequence here.				

Visualization

Drill

Slider A

Slider B

Counter

Counter 0

Errors 14

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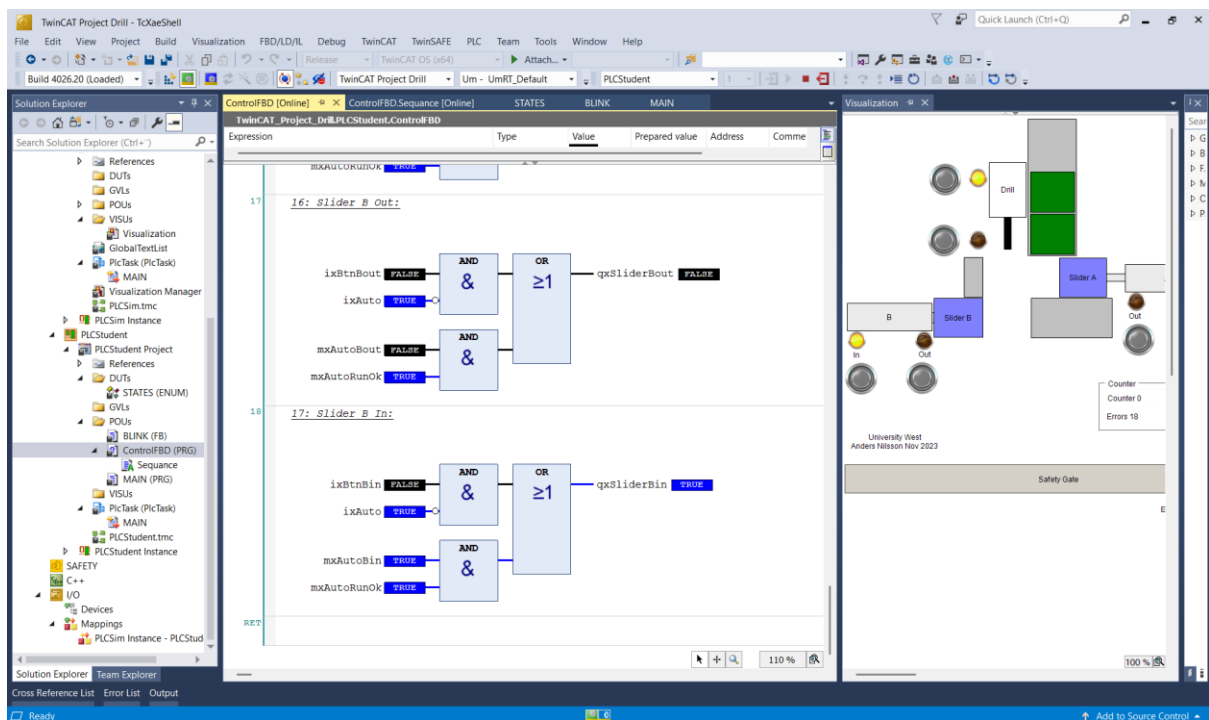
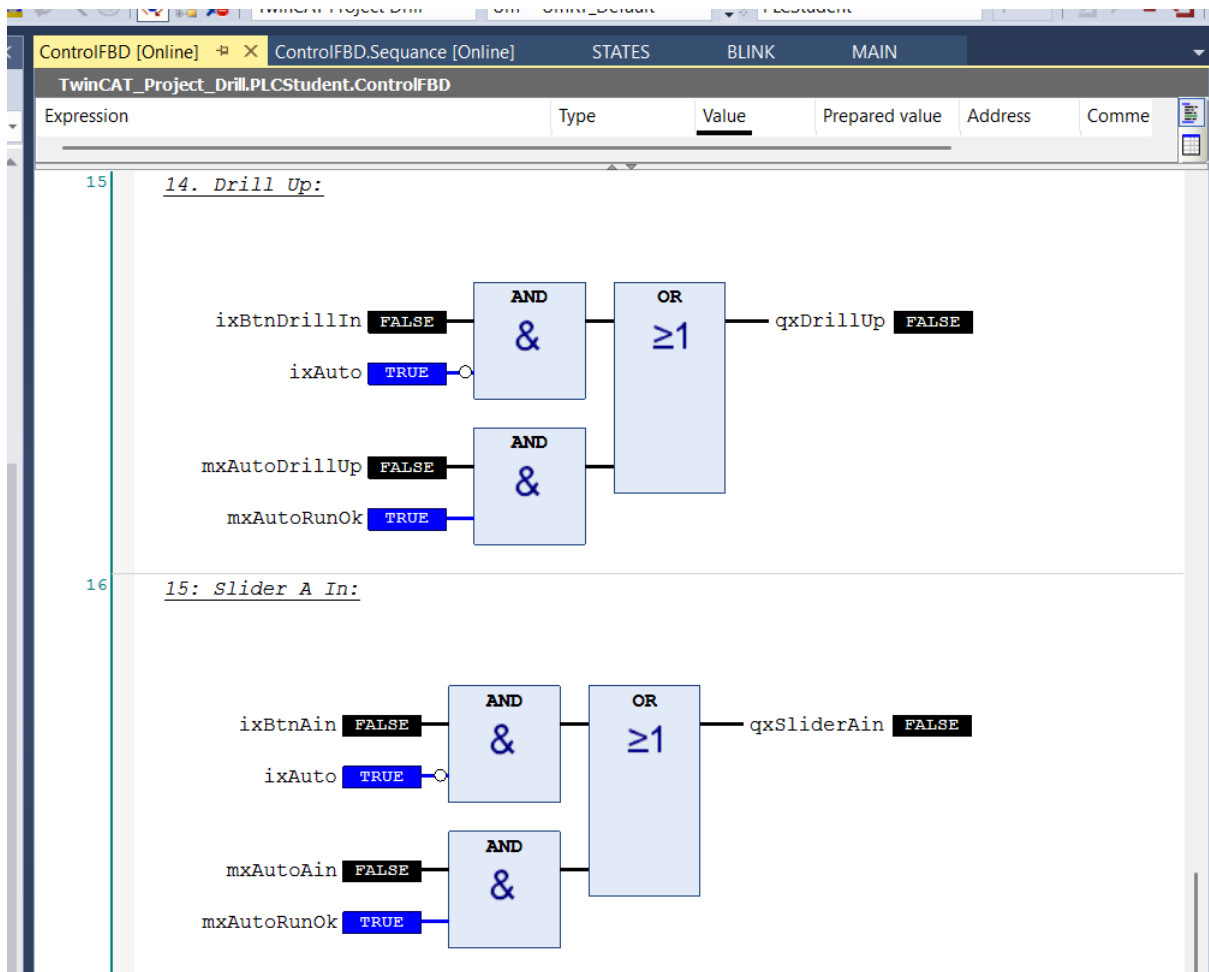
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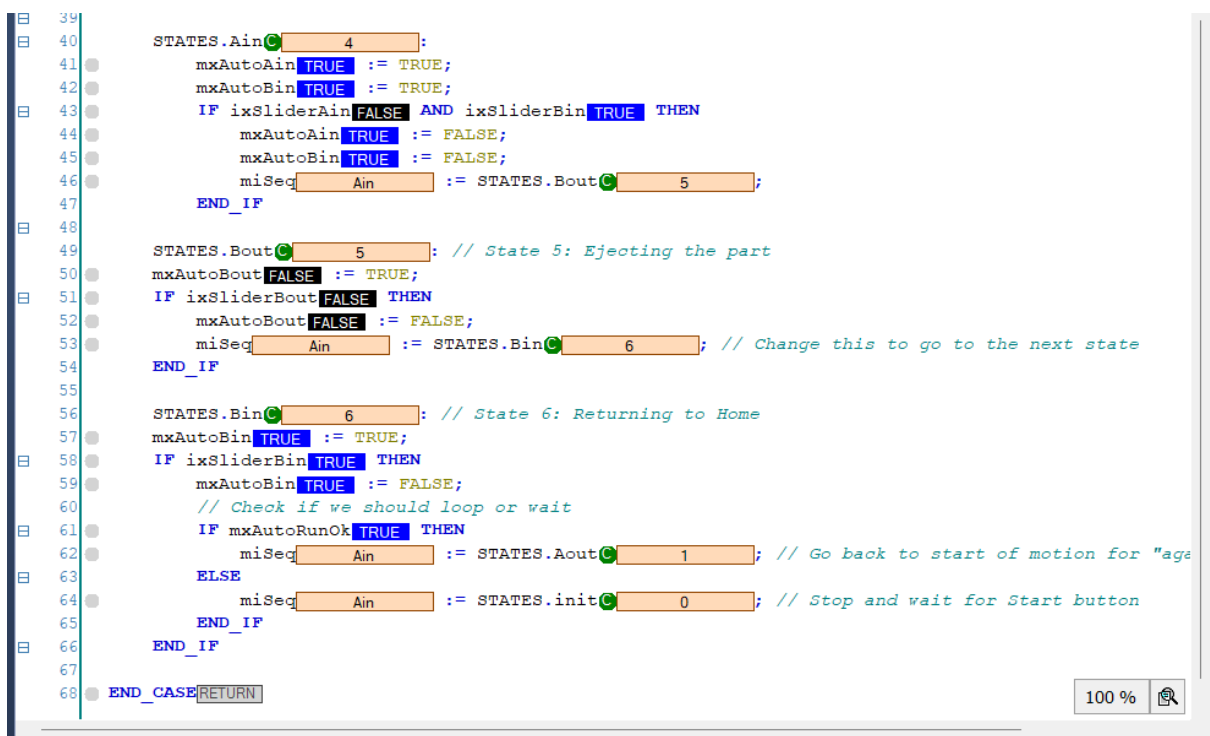
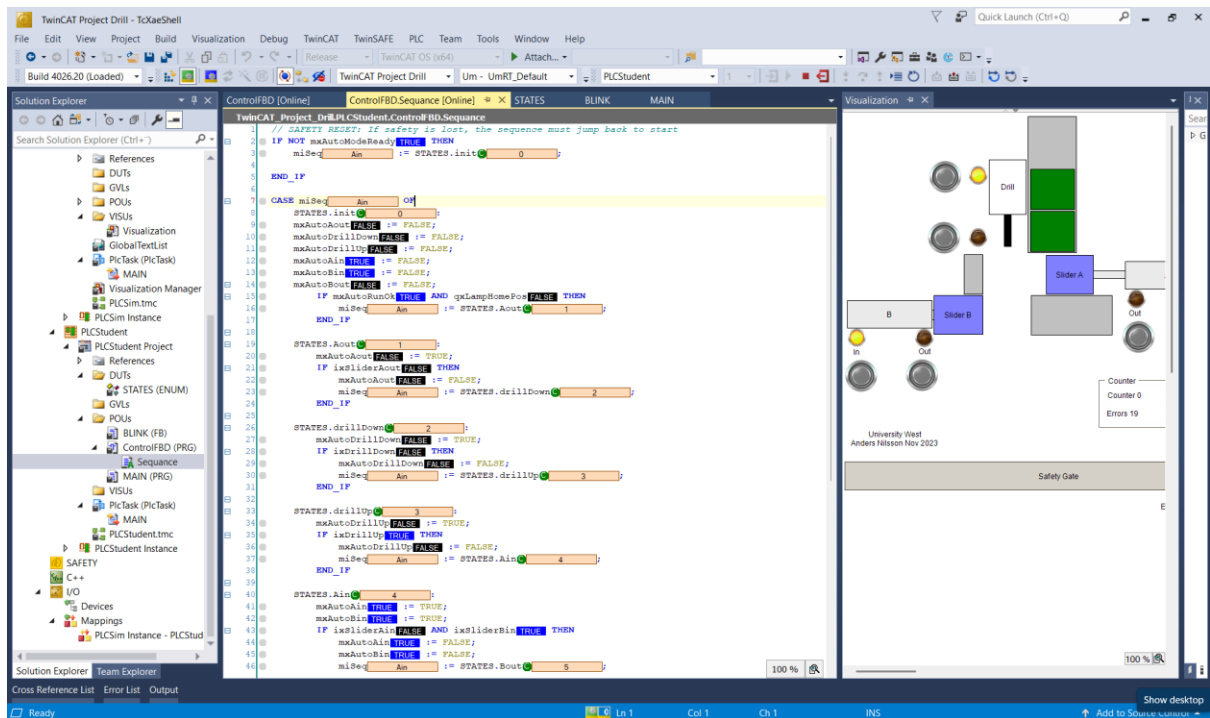
Safety Gate

ControlFBD [Online] ControlFBD.Sequence [Online] STATES BLINK MAIN

TwinCAT_Project_Drill.PLCStudent.ControlFBD

Expression	Type	Value	Prepared value	Address	Comme
13	<u>12. Create a network for slider A out (qxAuto):</u> We want the physical outputs (e.g. : qx) to turn on if EITHER the manual button is pressed OR the automate sequence commands it.				
14	<u>13. Create network for drill down:</u> qxDrillDown				





TwinCAT Project Drill - TCKaeShell

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BLINK [Online] ControlFBD [Online] ControlFBD Sequence [Online] STATES BLINK MAIN

TwinCAT_Project_Drill.PLCStudent.ControlFBD.FB.Blink

Expression	Type	Value	Prepared value	Address	Comment
OnTime	TIME	T#500ms			How long the
OffTime	TIME	T#500ms			How long the
OUT	BOOL	FALSE			The pulsing
fbTon	TON				Timer for ON
fbTof	TON				Timer for OF
mSeq	STATES	init			"m" for Men

```

1 | EfbTon (IN FALSE, I:= NOT EfbTof, Q FALSE, PT T#500ms, I:= OnTime T#500ms) ;
2 | EfbTof (I:= FB_TON, I:= fbTon, Q FALSE, PT T#500ms, I:= OffTime T#500ms) ;
3 | OUT FALSE := EfbTon, Q FB_TON ;
4 | RETURN ;

```

Visualization

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Safety Gate

Counter
Counter 0
Errors 22

100 %

Ready In 1 Col 1 Ch 1 INS Add to Source Control

TwinCAT Project Drill - TCKaeShell

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MAIN [Online] BLINK [Online] ControlFBD [Online] ControlFBD Sequence [Online] STATES

TwinCAT_Project_Drill.PLCStudent-MAIN

Expression	Type	Value	Prepared value	Address	Comment
ControlFBD ()	PRG				

```

1 | ControlFBD (); RETURN ;

```

Visualization

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Safety Gate

Counter
Counter 0
Errors 23

100 %

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